

Vectors and Vector Algebra

SS ILP Physics (Intermediate), Week 1 Class 1

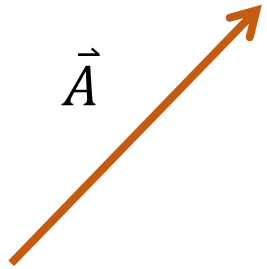


Outline

- Vectors
- Coordinate systems and vectors
- Vector Algebra

Vector

- A vector is a quantity that has magnitude and direction.
- A vector can be represented by a directed arrow in space.



- The arrow represents a vector denoted $\vec{A}$:
 1. Magnitude is $|\vec{A}| = A$
 2. Direction is $\vec{A}/A = \hat{A}$, where $\hat{A}$ is the **unit vector** of magnitude **1** parallel to $\vec{A}$.)
 3. Thus, mathematically $\vec{A} = A\hat{A}$.

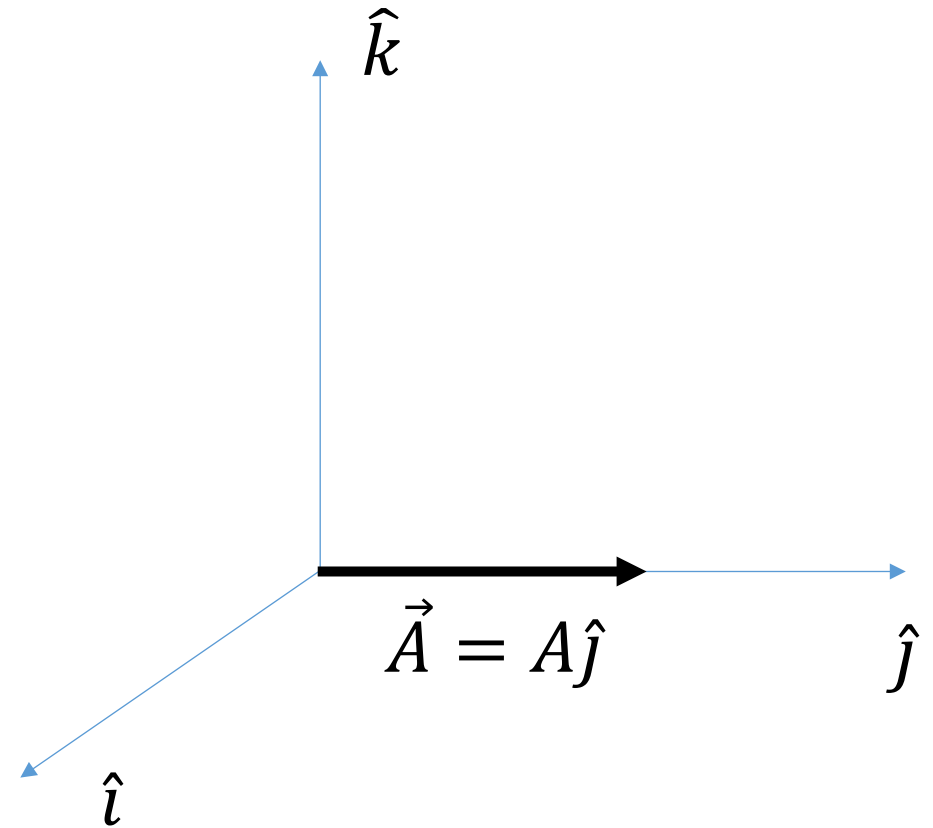
- 2 vectors are identical if and only if they have the same **magnitude** and **direction**.

$$\vec{A} = \vec{B} \Leftrightarrow |\vec{A}| = |\vec{B}| \text{ and } \hat{A} = \hat{B}.$$

<http://emweb.unl.edu/Math/mathweb/vectors/vectors.html>

Coordinate systems and vectors

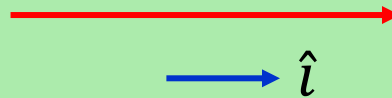
- Vectors can be conveniently represented in a coordinate system.
- With Cartesian coordinates, the positive direction on the x-axis is described by the unit vector $\hat{i}$, the positive direction on the y-axis by the unit vector $\hat{j}$, and the positive direction on the z-axis by the unit vector $\hat{k}$.



Example : A vector has magnitude of 5, and points in the direction of unit vector $\hat{i}$. What is the mathematical expression for the vector?

Ans : The vector is a product of magnitude and unit vector, i.e. $5\hat{i}$.

$$\vec{A} = A\hat{A} \quad A = 5 \text{ and } \hat{A} = \hat{i}$$



Coordinate systems and vectors

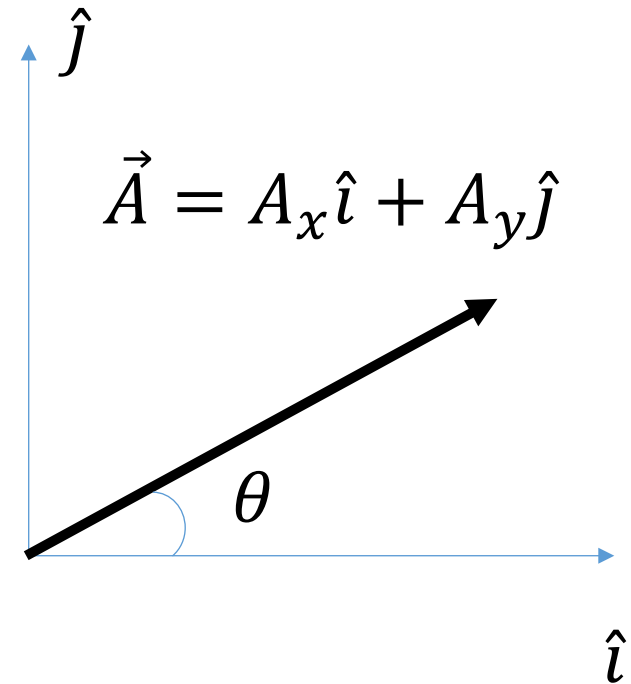
- Vectors can be conveniently represented in a coordinate system.
- Vectors in the x-y plane

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

$$A_x = A \cos \theta$$

$$A_y = A \sin \theta$$

where $A = \sqrt{A_x^2 + A_y^2}$ is the magnitude of $\vec{A}$.



Concept Question 1: Vector Quantity

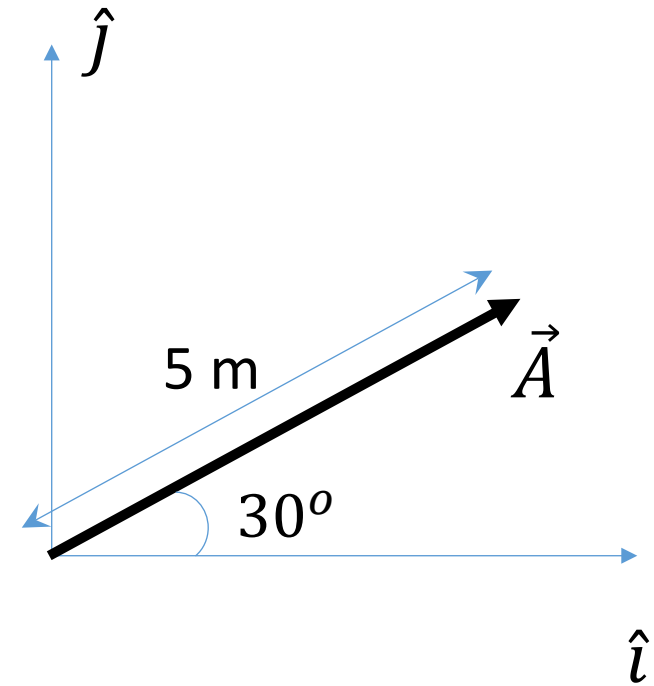
Given a vector graphically represented on the right, its coordinate expression is

A. $\vec{A} = 5 \cos 30^\circ \hat{i} + 5 \sin 30^\circ \hat{j}$.

B. $\vec{A} = \cos 30^\circ \hat{i} + \sin 30^\circ \hat{j}$.

C. $\vec{A} = 5 \sin 30^\circ \hat{i} + 5 \cos 30^\circ \hat{j}$.

D. $\vec{A} = \sin 30^\circ \hat{i} + \cos 30^\circ \hat{j}$.



Concept Question 1: Solution

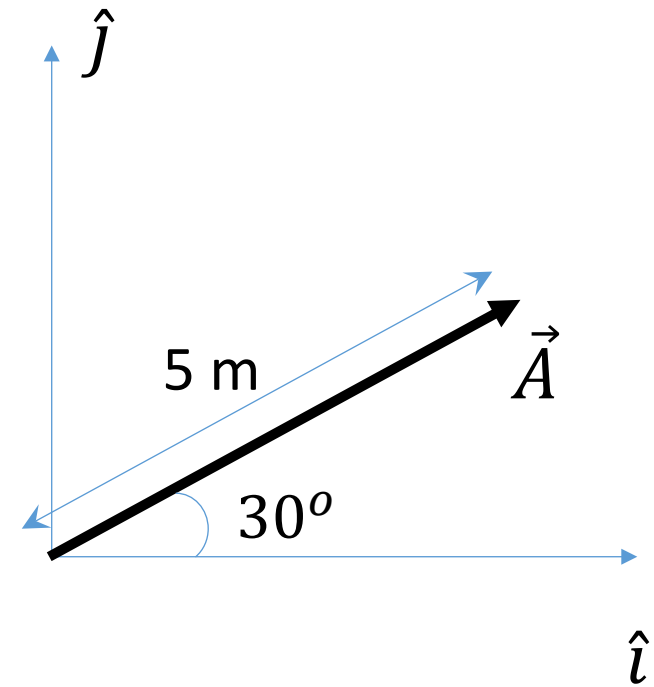
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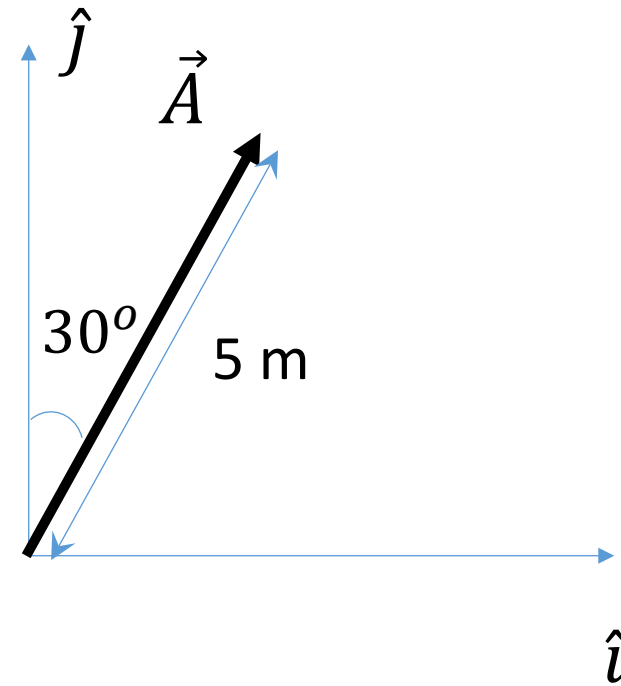


Concept Question 2: Vector Quantity

Given a vector graphically represented on the right, its coordinate expression is

A. $\vec{A} = 5 \cos 30^\circ \hat{i} + 5 \sin 30^\circ \hat{j}$.

B. $\vec{A} = 5 \cos 60^\circ \hat{i} + 5 \sin 60^\circ \hat{j}$.

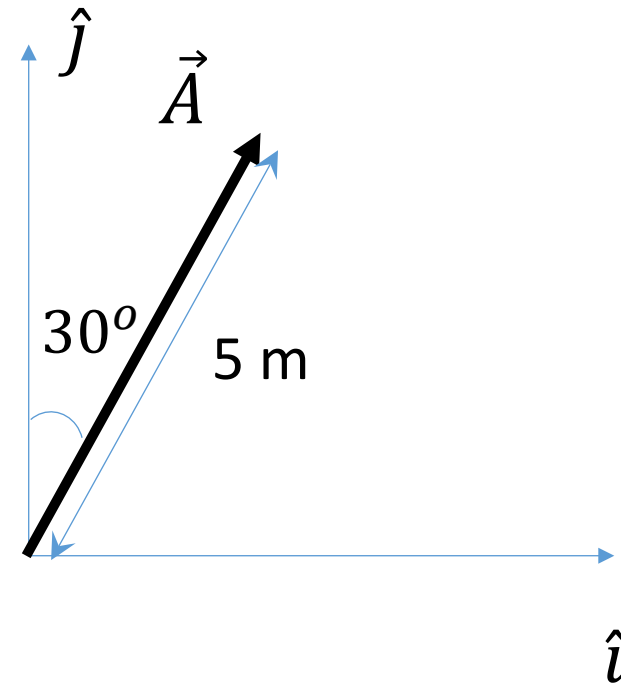


Concept Question 2: Solution

Given a vector graphically represented on the right, its coordinate expression is

A. $\vec{A} = 5 \cos 30^\circ \hat{i} + 5 \sin 30^\circ \hat{j}$.

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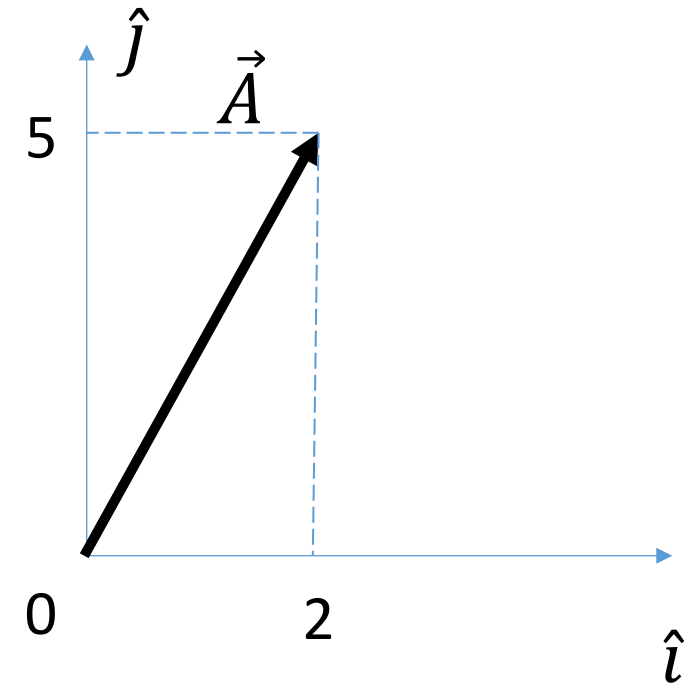


Concept Question 3: Vector Quantity

Given a vector graphically represented on the right, its coordinate expression is

A. $\vec{A} = 5 \hat{i} + 2 \hat{j}$.

B. $\vec{A} = 2 \hat{i} + 5 \hat{j}$.

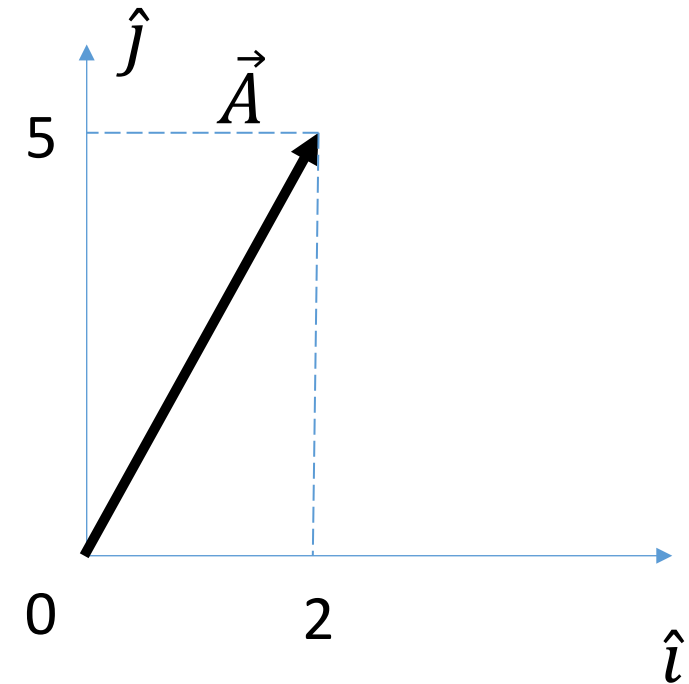


Concept Question 3: Solution

Given a vector graphically represented on the right, its coordinate expression is

A. $\vec{A} = 5 \hat{i} + 2 \hat{j}$.

B. $\vec{A} = 2 \hat{i} + 5 \hat{j}$.



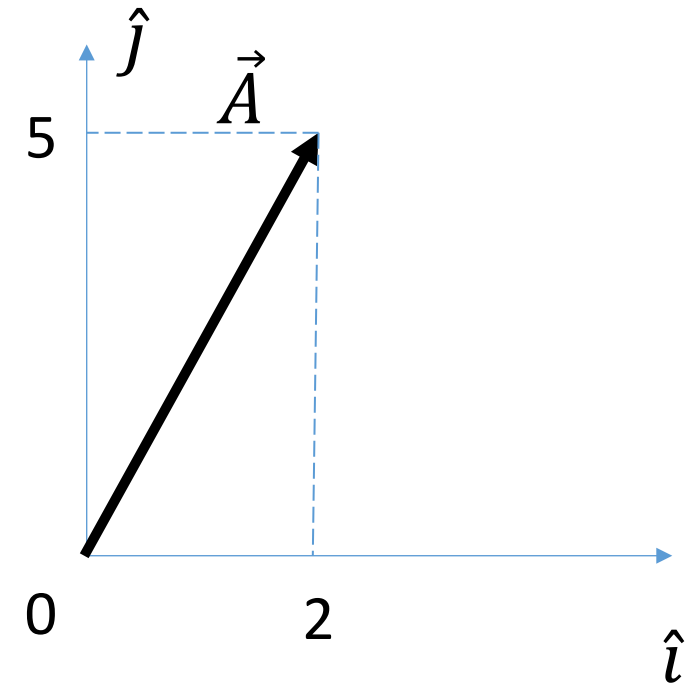
Concept Question 4: Vector Quantity

Given a vector graphically represented on the right, its magnitude is

A. $A = 5$.

B. $A = 2$.

C. $A = \sqrt{5^2 + 2^2}$.



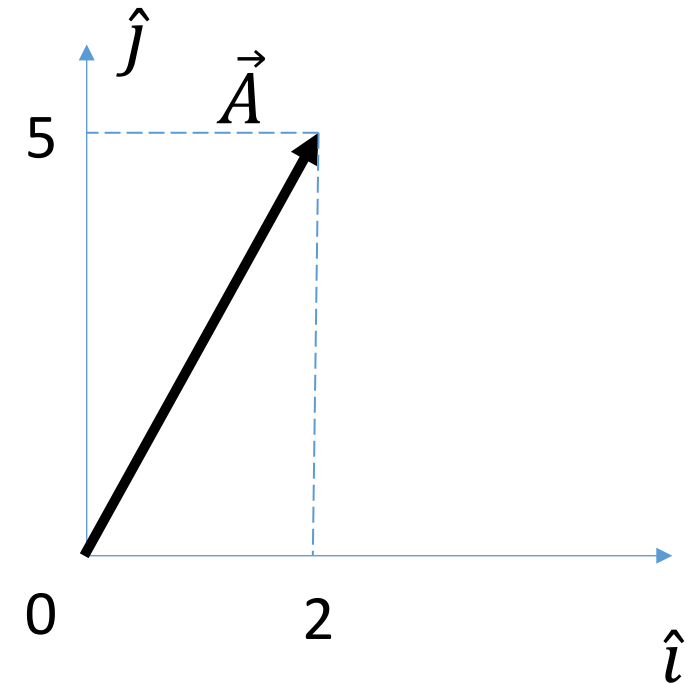
Concept Question 4: Vector Quantity

Given a vector graphically represented on the right, its magnitude is

A. $A = 5$.

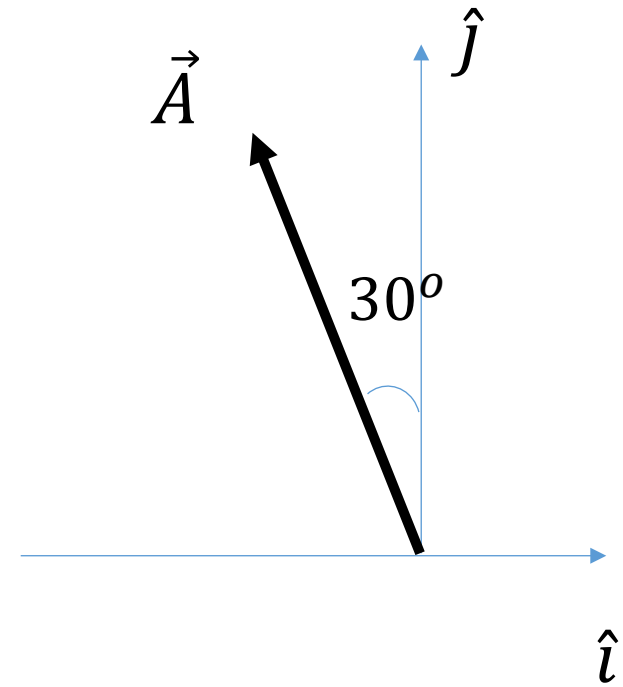
B. $A = 2$.

C. $A = \sqrt{5^2 + 2^2}$.



Case problem 1

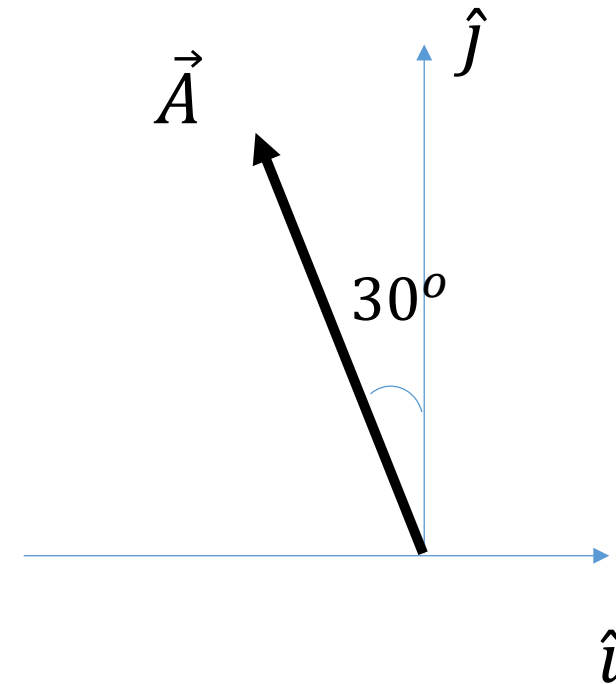
Given a vector of magnitude 5 graphically represented on the right, find its coordinate expression.



Case problem 1: Solution

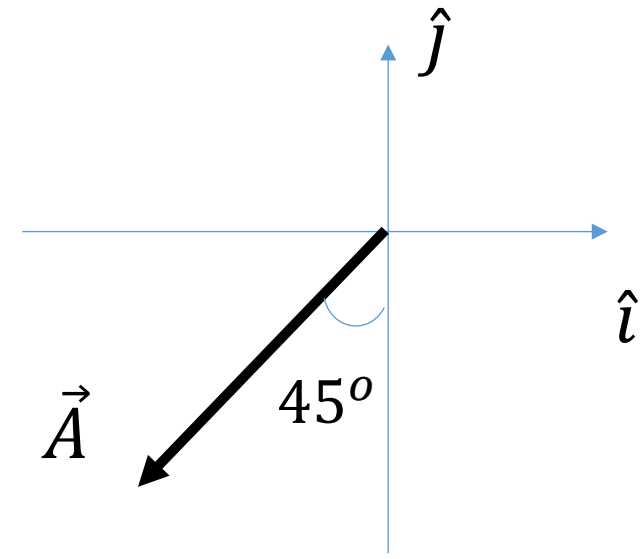
Given a vector of magnitude 5 graphically represented on the right, find its coordinate expression.

$$\vec{A} = 5 \cos 60^\circ (-\hat{i}) + 5 \sin 60^\circ \hat{j}$$



Case problem 2

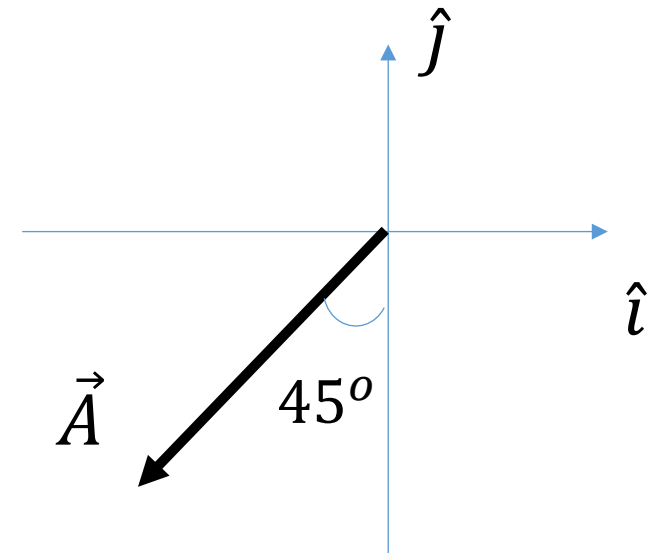
Given a vector of magnitude 3 graphically represented on the right, find its coordinate expression.



Case problem 2: Solution

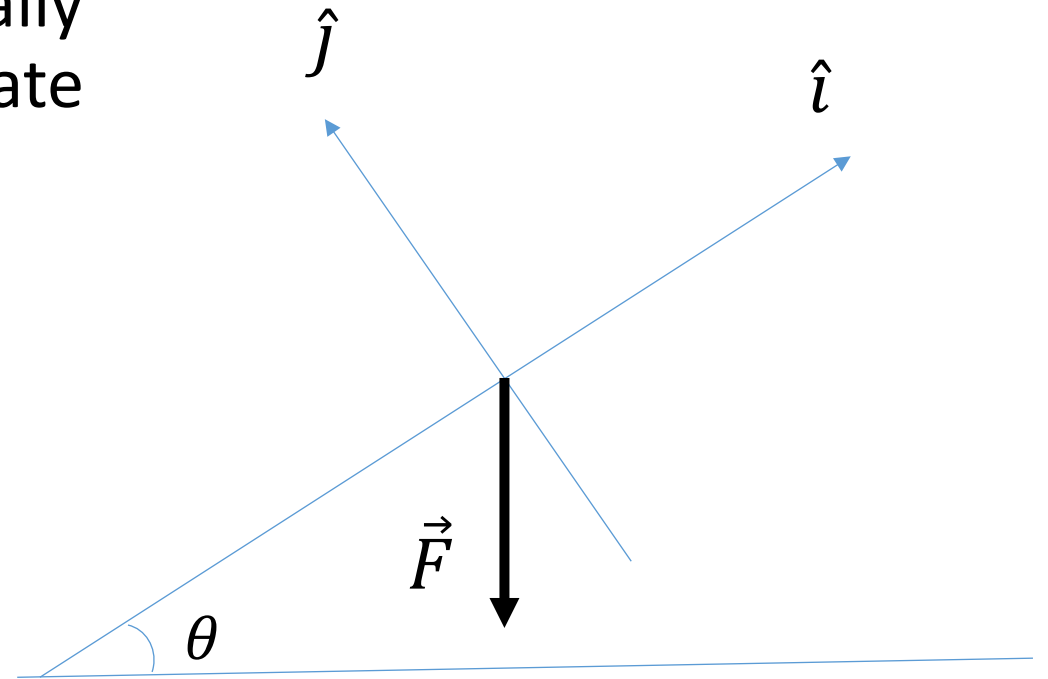
Given a vector of magnitude 3 graphically represented on the right, find its coordinate expression.

$$\vec{A} = 3 \cos 45^\circ (-\hat{i}) + 3 \sin 45^\circ (-\hat{j})$$



Case problem 3

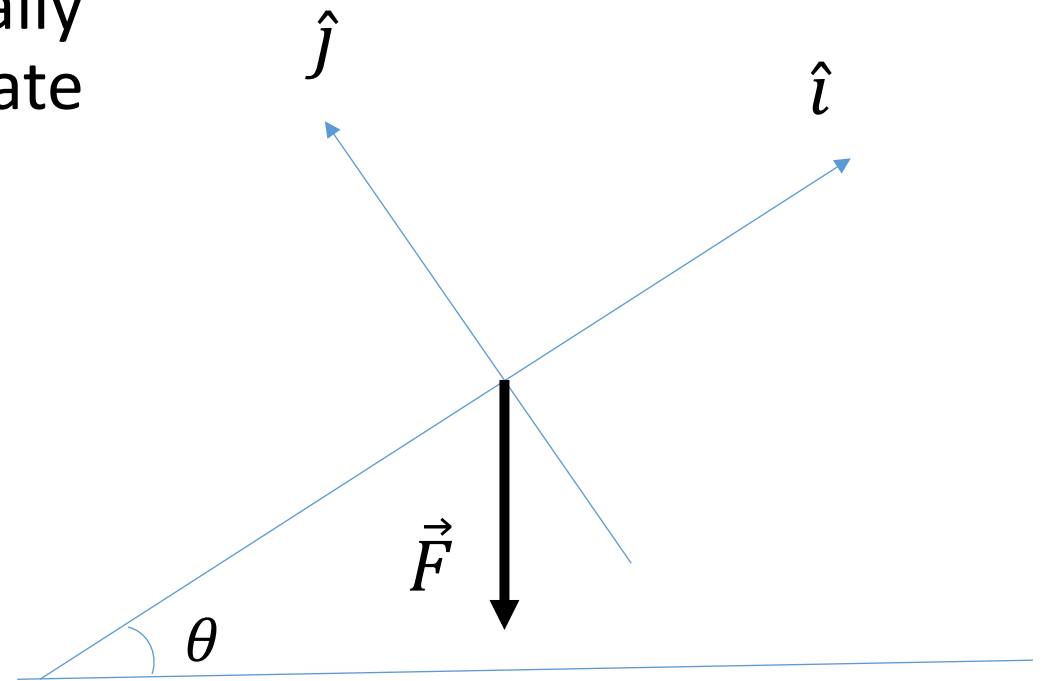
Given a vector of magnitude F graphically represented on the right, find its coordinate expression.



Case problem 3: Solution

Given a vector of magnitude F graphically represented on the right, find its coordinate expression.

$$\vec{F} = F \sin \theta (-\hat{i}) + F \cos \theta (-\hat{j})$$



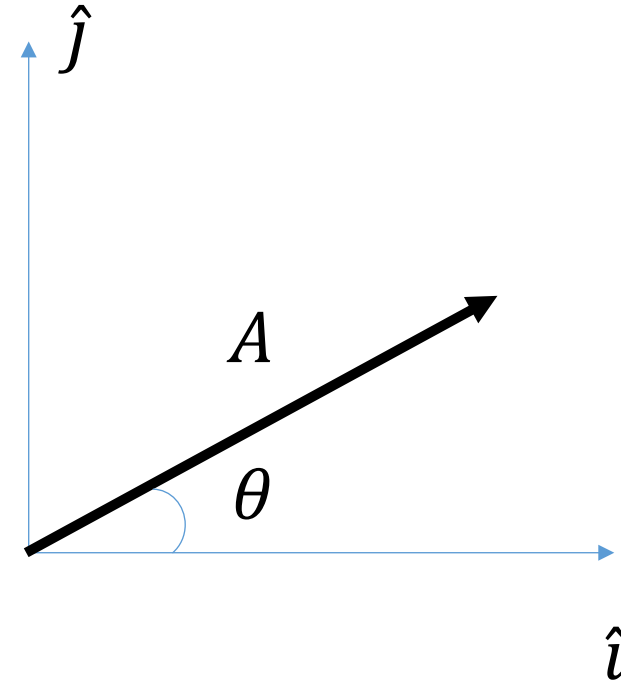
Coordinate systems and vectors

Given the magnitude A and angle θ of $\vec{A}$, we can find

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

where $A_x = A \cos \theta$

$$A_y = A \sin \theta$$

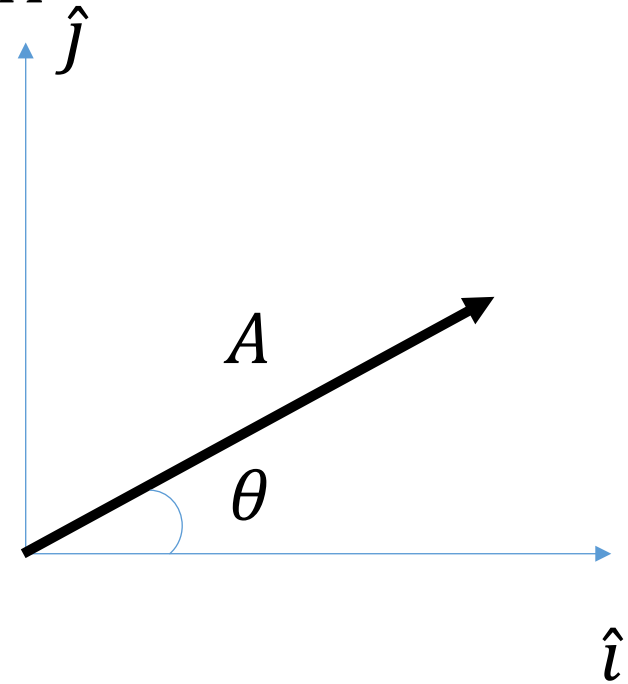


Coordinate systems and vectors

Given $\vec{A} = A_x \hat{i} + A_y \hat{j}$, we can find the magnitude A and angle θ of $\vec{A}$ as

$$A = \sqrt{A_x^2 + A_y^2}$$

$$\theta = \tan^{-1} \frac{A_y}{A_x}$$



Concept question 5: Vector Quantity

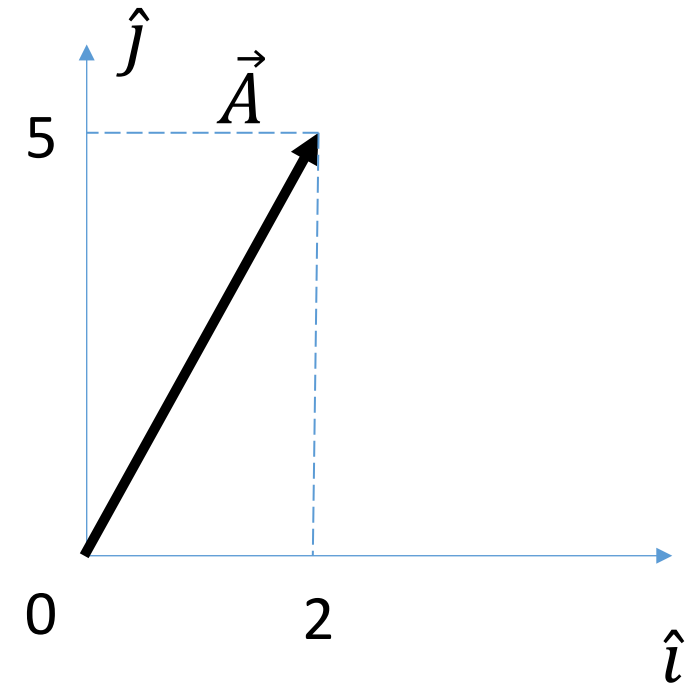
Given a vector graphically represented on the right, its angle is

A. $\theta = \tan^{-1} \frac{5}{2}.$

B. $\theta = \tan^{-1} \frac{2}{5}.$

C. $\theta = \pi - \tan^{-1} \frac{5}{2}.$

D. $\theta = \pi - \tan^{-1} \frac{2}{5}.$



Concept question 5: Solution

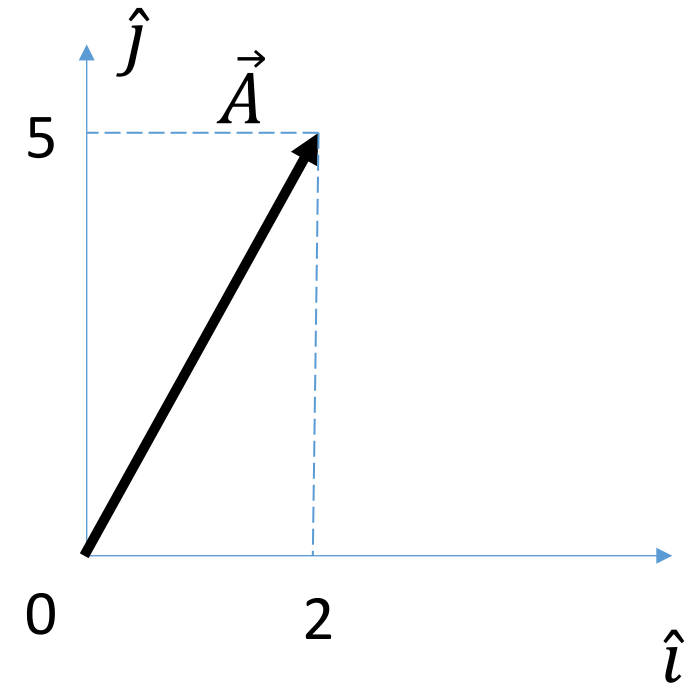
Given a vector graphically represented on the right, its angle is

A. $\theta = \tan^{-1} \frac{5}{2}.$

B. $\theta = \tan^{-1} \frac{2}{5}.$

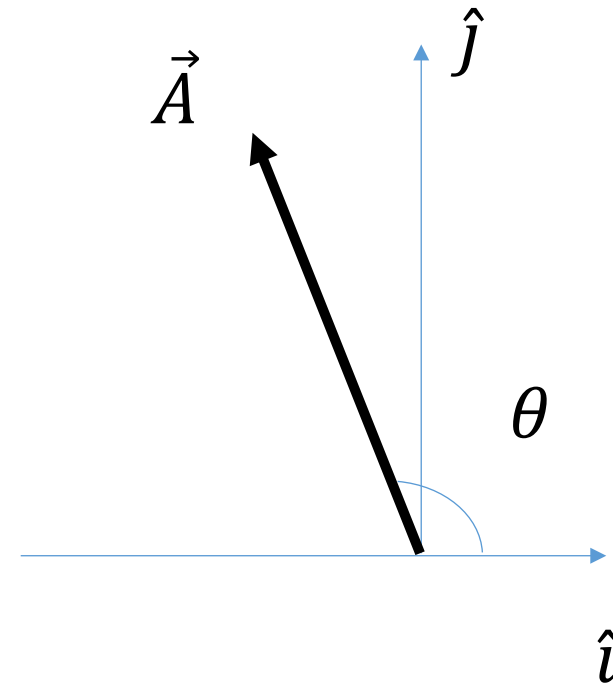
C. $\theta = \pi - \tan^{-1} \frac{5}{2}.$

D. $\theta = \pi - \tan^{-1} \frac{2}{5}.$



Case problem 4

Given a vector $\vec{A} = -3\hat{i} + 4\hat{j}$ which is graphically represented on the right, find its magnitude A and angle θ .

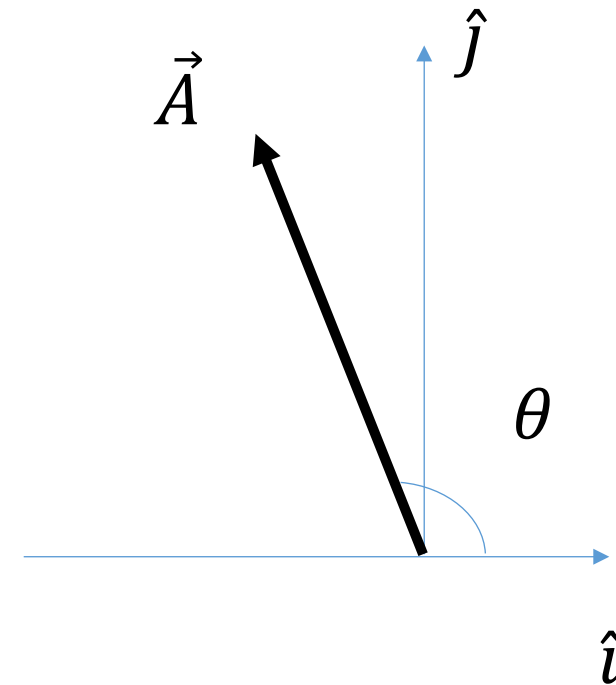


Case problem 4: Solution

Given a vector $\vec{A} = -3\hat{i} + 4\hat{j}$ which is graphically represented on the right, find its magnitude A and angle θ .

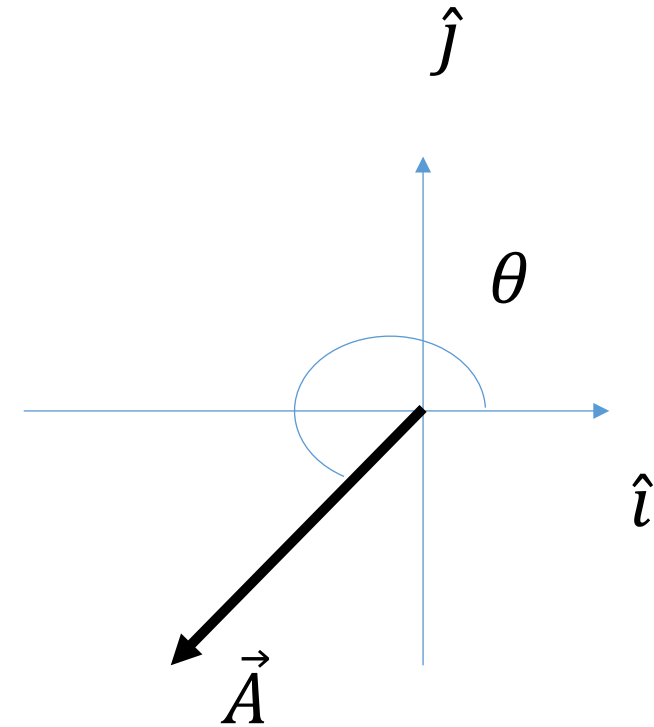
$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{(-3)^2 + 4^2} = 5$$

$$\theta = \pi + \tan^{-1} \frac{A_y}{A_x} = \pi - \tan^{-1} \frac{4}{3}$$



Case problem 5

Given a vector $\vec{A} = -3\hat{i} - 3\hat{j}$ which is graphically represented on the right, find its magnitude A and angle θ .

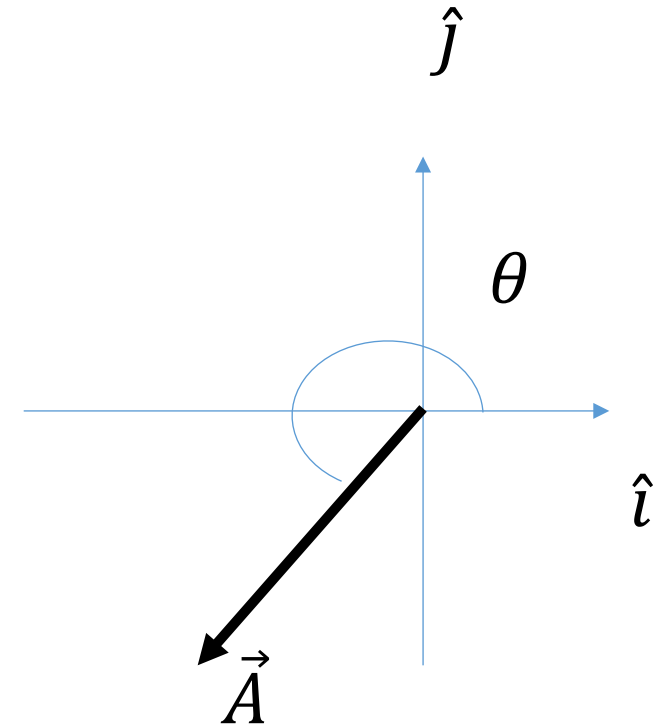


Case problem 5: Solution

Given a vector $\vec{A} = -3\hat{i} - 3\hat{j}$ which is graphically represented on the right, find its magnitude A and angle θ .

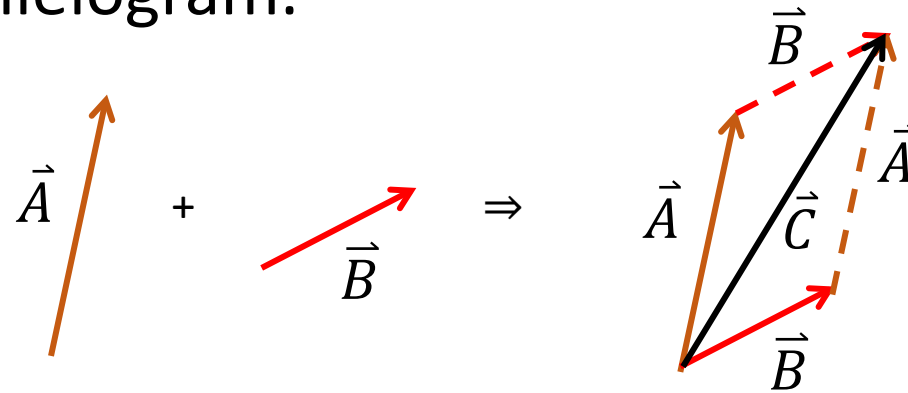
$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{(-3)^2 + (-3)^2} = 3\sqrt{2}$$

$$\theta = \pi + \tan^{-1} \frac{A_y}{A_x} = \pi + \tan^{-1} 1 = \frac{5\pi}{4}$$



Vector Algebra

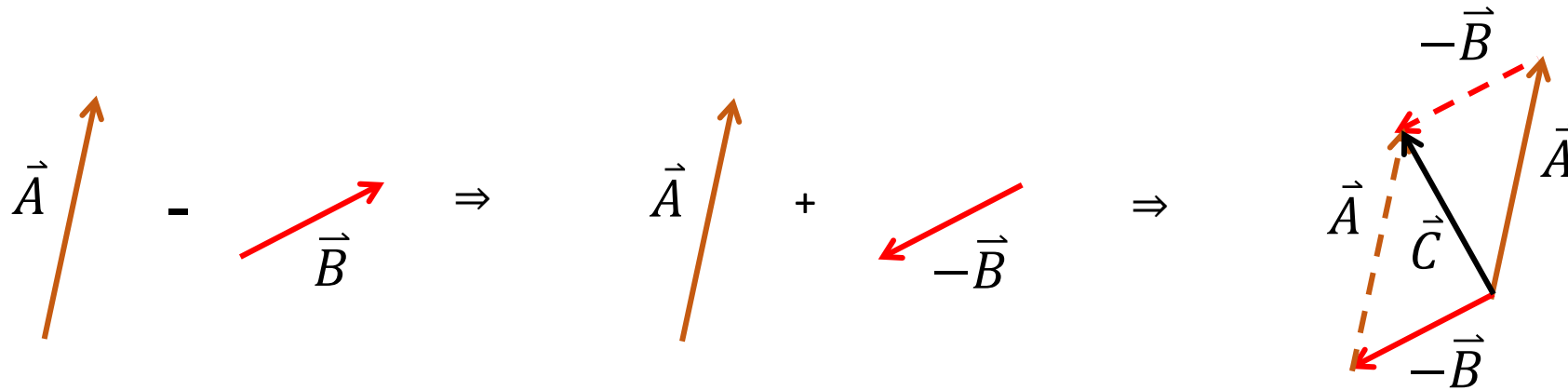
- Vector summation : $\vec{A} + \vec{B} = \vec{C}$
 1. Parallely move the 2 vectors $\vec{A}$ and $\vec{B}$ until their tails meet.
 2. Construct a parallelogram as shown.
 3. The sum of $\vec{A}$ and $\vec{B}$ is the vector $\vec{C}$ originated from the tails along the diagonal of parallelogram.



<http://emweb.unl.edu/Math/mathweb/vectors/vectors.html>

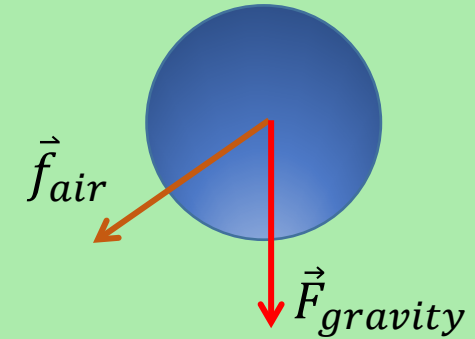
Vector Algebra

- Vector subtraction : $\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) = \vec{C}$,
 1. Construct new vector $-\vec{B}$.
 2. Parallely move $\vec{A}$ and $-\vec{B}$ until their tails meet.
 3. Construct a parallelogram.
 4. The sum of $\vec{A}$ and $-\vec{B}$ or the subtraction of $\vec{A}$ and $\vec{B}$ is the vector $\vec{C}$ originated from the tails in the diagonal direction.

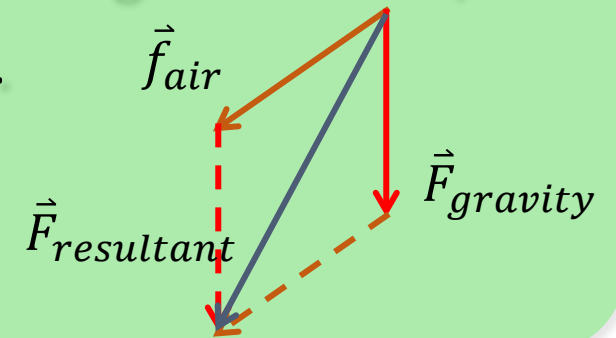


Vector algebra in physics

Example : A ball is kicked out and experiences air resistance and gravitational force as shown. Determine the resultant force by vector algebra.



Ans : The net resultant force is the vector sum of the 2 forces acting on the ball, $\vec{F}_{resultant} = \vec{f}_{air} + \vec{F}_{gravity}$ and can be constructed as shown.



Coordinate systems and vector algebra

Given two vectors $\vec{A} = A_x\hat{i} + A_y\hat{j}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j}$,

$$\vec{A} + \vec{B} = A_x\hat{i} + A_y\hat{j} + B_x\hat{i} + B_y\hat{j} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j}$$

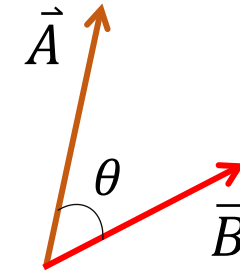
$$\vec{A} - \vec{B} = A_x\hat{i} + A_y\hat{j} - (B_x\hat{i} + B_y\hat{j}) = (A_x - B_x)\hat{i} + (A_y - B_y)\hat{j}$$

Vector Dot Product (Scalar Product)

- Vector dot product of $\vec{A}$ and $\vec{B}$ is defined as

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

where θ is the angle formed by the 2 vectors.



- Dot product of 2 vectors is a pure **scalar** (a quantity that has magnitude but no direction).
- Dot product is commutative, i.e. $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$
- Dot product is distributive, i.e.

$$(\vec{A} + \vec{B}) \cdot (\vec{C} + \vec{D}) = \vec{A} \cdot \vec{C} + \vec{A} \cdot \vec{D} + \vec{B} \cdot \vec{C} + \vec{B} \cdot \vec{D}$$

<http://emweb.unl.edu/Math/mathweb/vectors/vectors.html>

Coordinate systems and vector algebra

Given two vectors $\vec{A} = A_x \hat{i} + A_y \hat{j}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j}$,

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$$

Dot Product in Physics

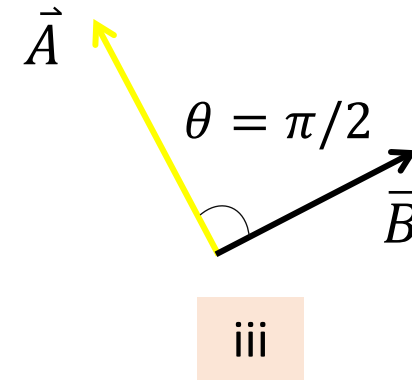
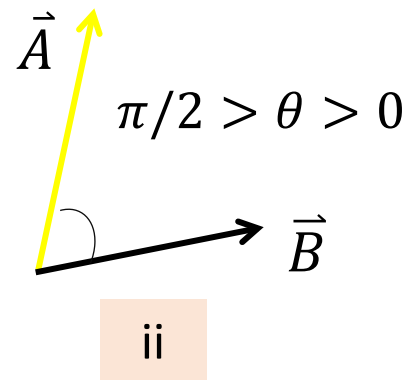
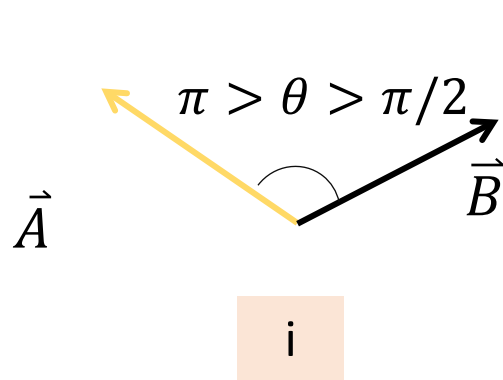
- Work done W by a force $\vec{F}$ while displacing an object by displacement $\vec{S}$,
 $W = \vec{F} \cdot \vec{S}$
- Maxwell's equations

[https://en.wikipedia.org/wiki/Work_\(physics\)](https://en.wikipedia.org/wiki/Work_(physics))

https://en.wikipedia.org/wiki/Maxwell%27s_equations

Concept Question 6: Vector Dot Product

Compare $\vec{A} \cdot \vec{B}$ in the following three cases.



- A. $i > ii > iii$
- B. $i > iii > ii$
- C. $ii > i > iii$
- D. $ii > iii > i$

Concept Question 6: Solution

Ans : D. ii > iii > i

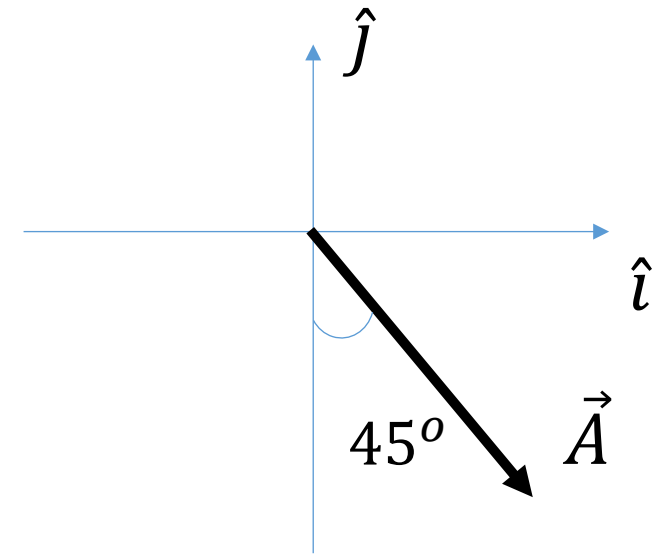
The sign of the dot product depends on the angle, which determines the sign of the cosine function.

$$\begin{array}{ll} 1 \geq \cos \theta > 0 & \text{for } \pi/2 > \theta \geq 0 \\ 0 > \cos \theta \geq -1 & \text{for } \pi \geq \theta > \pi/2 \\ \cos \theta = 0 & \text{for } \theta = \pi/2 \end{array}$$

Off-class practice question(s)

Case problem

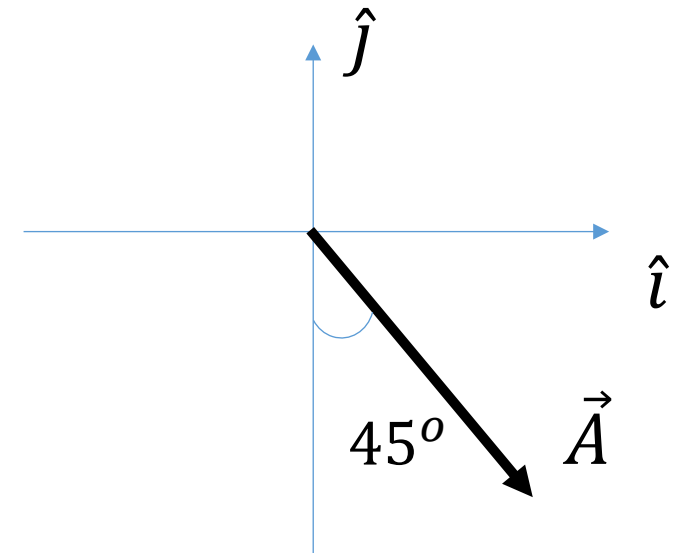
Given a vector of magnitude 3 graphically represented on the right, find its coordinate expression.



Case problem: Solution

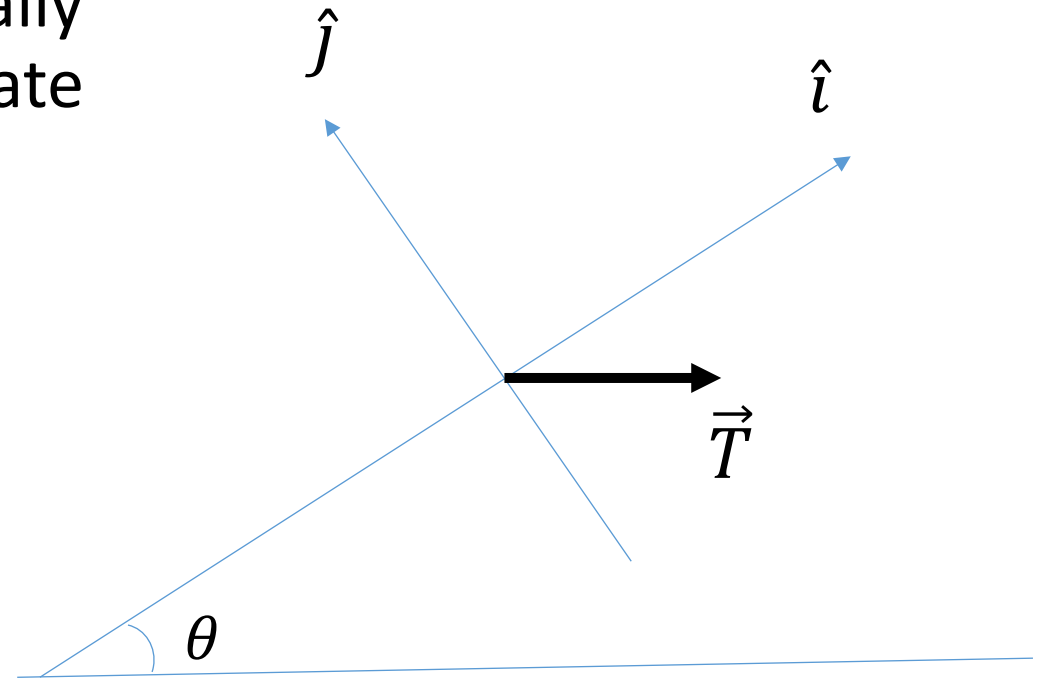
Given a vector of magnitude 3 graphically represented on the right, find its coordinate expression.

$$\vec{A} = 3 \cos 45^\circ \hat{i} + 3 \sin 45^\circ (-\hat{j})$$



Case problem

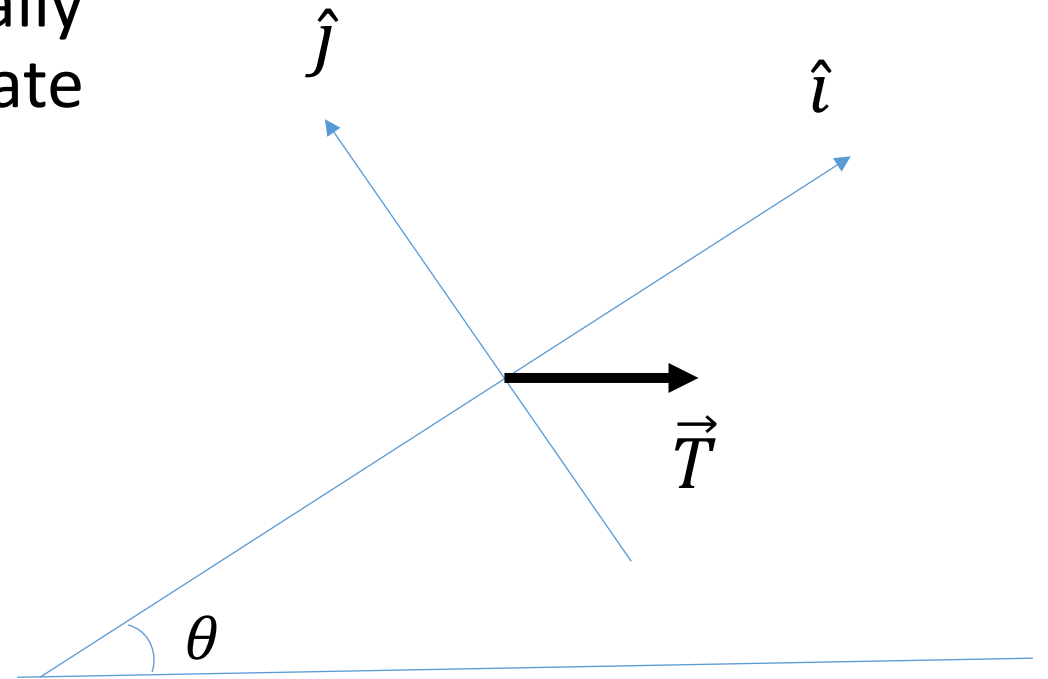
Given a vector of magnitude T graphically represented on the right, find its coordinate expression.



Case problem: Solution

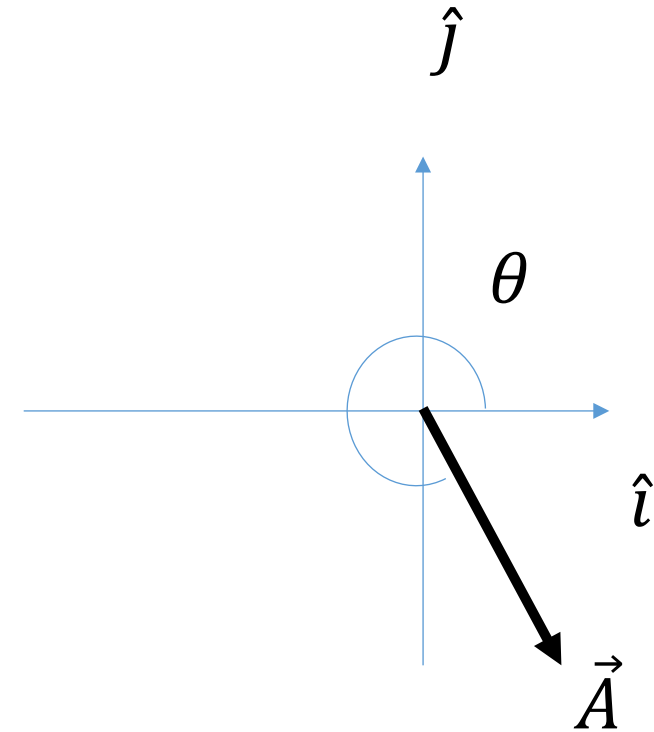
Given a vector of magnitude T graphically represented on the right, find its coordinate expression.

$$\vec{T} = T \cos \theta \hat{i} + T \sin \theta (-\hat{j})$$



Case problem

Given a vector $\vec{A} = \hat{i} - \sqrt{3}\hat{j}$ which is graphically represented on the right, find its magnitude A and angle θ .

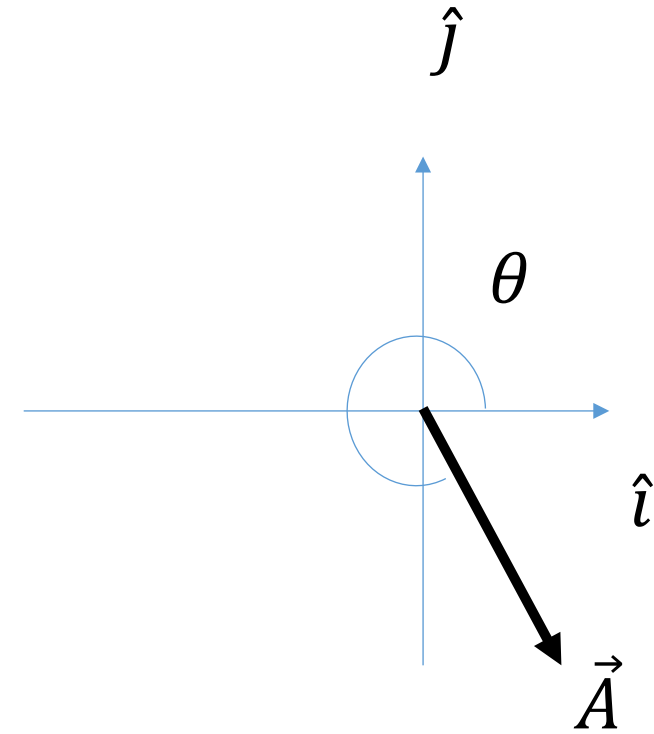


Case problem: Solution

Given a vector $\vec{A} = \hat{i} - \sqrt{3}\hat{j}$ which is graphically represented on the right, find its magnitude A and angle θ .

$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{(1)^2 + (-\sqrt{3})^2} = 2$$

$$\theta = 2\pi + \tan^{-1} \frac{A_y}{A_x} = 2\pi - \tan^{-1} \sqrt{3} = \frac{5\pi}{3}$$



Concept Question: Vector Algebra

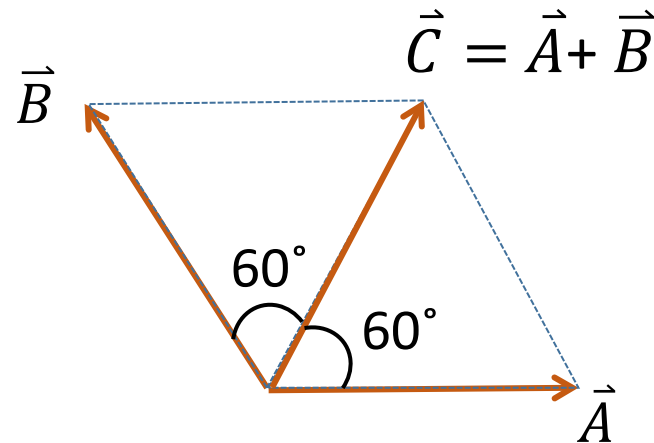
Which of the following statements is correct?

- A. The sum of 2 vectors of equal length that are 45° apart gives a vector of the same length as them.
- B. The sum of 2 vectors of equal length that are 60° apart gives a vector of the same length as them.
- C. The sum of 2 vectors of equal length that are 90° apart gives a vector of the same length as them.
- D. The sum of 2 vectors of equal length that are 120° apart gives a vector of the same length as them.
- E. The sum of 2 vectors is always larger in magnitude than each of the 2 vectors.

Concept Question: Solution

Ans : D. The sum of 2 vectors of equal length that are 120° apart gives a vector of the same length as them.

Following the rule of vector sum construction, given $|\vec{A}| = |\vec{B}|$,



$$|\vec{C}| = |\vec{A}| = |\vec{B}|$$